

Roll No.

TPE-717

**B. TECH. (PE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

ADVANCED EOR TECHNIQUE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Explain the different oil production phases. Why do we need EOR ?
Currently available EOR is based on which parameter ?

10 Marks (CO1)

OR

- (b) Discuss the methods of EOR. Explain any *one* method with schematic diagram for EOR.

10 Marks (CO1)

2. (a) Draw the schematic diagram of MEOR and explain the each step in increasing the production rate.

10 Marks (CO2)

OR

- (b) Write and define the different types of oil field pressure calculation requirements.

10 Marks (CO2)

3. (a) Define pressure gradient in the well. Also explain the mathematical expression for the hydrocarbon interface (H_c) of gas-oil (GOC), oil-water (OWC). Specify the symbol used.

10 Marks (CO2)

P. T. O.

OR

(b) The data of depth and pressure is given in the below :

Depth (m)	Pressure (atm.)
0	51.6
200	55.2
400	59.2
600	63.8
800	81.4
1000	99.0
1200	119.0
1400	140.0

Calculate the following :

- (i) Pressure gradients
- (ii) Depth of gas-oil and gas-water interface
- (iii) Pressures at GOC and WOC

Given : Depth of top perforation = 1150 m, GOC = 740 m,

WOC = 1170 m.

10 Marks (CO2)

4. (a) What are the factors on which the decline curves are characterized ?

Write down the first condition that must be considered in production-decline-curve analysis.

10 Marks (CO1/CO2)

(3)

OR

- (b) Define ARPs equation. Discuss the types of rate decline behaviour in the estimation of reserves. Write with mathematical expression.

10 Marks (CO1/CO2)

5. (a) The following production data are available from a dry gas field :

q_t (MMscf/D)	Gp (MMscf)	q_t (MMscf/D)	Gp (MMscf)
320	16000	208	304000
336	32000	197	352000
304	48000	184	368000
309	96000	176	384000
272	160000	184	400000

Estimate the cumulative gas production when the gas flow rate reaches 80 MMscf/D.

10 Marks (CO2)

OR

- (b) Discuss the different selection patterns of water flooding. Discuss the following in water flooding during EOR :

10 Marks (CO2)

- (i) Overall recovery efficiency
- (ii) Displacement efficiency
- (iii) Volumetric sweep efficiency